Charlotte MLB Analysis

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Loading Packages and Dataset

```r
library(Lahman)
library(readxl)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(ggplot2)
```

```r
teams <- read_excel("BaseballAttendanceData.xlsx")
```

```r
teams2 <- Teams
```

Data Cleaning

```r
teams <- na.omit(teams)
teams$win_pct <- as.numeric(teams$win_pct)
head(teams)
```

```
## # A tibble: 6 x 12
##    rank yearID franchID attendance  Wins Games HomeGames win_pct population
##   <dbl>  <dbl> <chr>         <dbl> <dbl> <dbl>     <dbl>   <dbl>      <dbl>
## 1     1   2000 ANA         2066982    82   162        81   0.506   11798000
## 2     2   2001 ANA         2000919    75   162        81   0.463   11834000
## 3     3   2002 ANA         2305547    99   162        81   0.611   11870000
## 4     4   2003 ANA         3061094    77   162        82   0.475   11906000
## 5     5   2004 ANA         3375677    92   162        81   0.568   11942000
## 6     6   2005 ANA         3404686    95   162        81   0.586   11978000
## # i 3 more variables: NBAandNFL <dbl>, MedHouseIncome <dbl>, teamPayroll <dbl>
```

```r
teams2 <- teams2 %>%
  filter(yearID >= 2000 & yearID <= 2019) %>%
  select(yearID, franchID, divID, Rank, DivWin, WSWin, HR, SHO)


teams <- left_join(teams, teams2, by = c("yearID", "franchID"))
```

a)

Linear regression model for attendance

```r
attendanceModel <- lm(attendance ~ population + NBAandNFL + MedHouseIncome + teamPayroll + win_pct + as
```

> For the first linear model, we assess attendance as a function of population, number of NFL and NBA teams in the same location, win percentage of the season, median household income of the area, the teams payroll, the teams division, team rank, World Series win, home runs, and shut outs.

Summary of the model

```r
summary(attendanceModel)
```

```
##
## Call:
## lm(formula = attendance ~ population + NBAandNFL + MedHouseIncome +
##     teamPayroll + win_pct + as.factor(Rank) + as.factor(WSWin) +
##     HR + SHO, data = teams)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1353922  -365681   -31233   372250  1282650
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)        8.318e+05  3.860e+05   2.155 0.031596 *
## population         1.851e-02  9.234e-03   2.005 0.045439 *
## NBAandNFL         -5.752e+03  4.412e+04  -0.130 0.896319
## MedHouseIncome    -4.411e+00  1.534e+00  -2.875 0.004189 **
## teamPayroll        7.724e-03  5.859e-04  13.184  < 2e-16 ***
## win_pct            3.507e+06  6.910e+05   5.076 5.27e-07 ***
## as.factor(Rank)2  -7.300e+04  7.001e+04  -1.043 0.297513
## as.factor(Rank)3  -1.158e+05  8.453e+04  -1.369 0.171428
## as.factor(Rank)4  -9.524e+04  1.078e+05  -0.884 0.377256
## as.factor(Rank)5  -1.975e+05  1.298e+05  -1.521 0.128840
## as.factor(Rank)6   1.263e+05  1.925e+05   0.656 0.511853
## as.factor(WSWin)Y -8.684e+04  1.178e+05  -0.737 0.461162
```

```
## HR                -2.471e+03  6.985e+02  -3.538 0.000437 ***
## SHO               -2.250e+04  6.742e+03  -3.338 0.000902 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 497800 on 558 degrees of freedom
## Multiple R-squared:  0.5019, Adjusted R-squared:  0.4903
## F-statistic: 43.25 on 13 and 558 DF,  p-value: < 2.2e-16
```

From the model summary, all variables are significant except for NBA and NFL teams, team rank, and World Series win. We will remove those variables.

```
attendanceModel <- lm(attendance ~ population + MedHouseIncome + teamPayroll + win_pct + HR + SHO, data

summary(attendanceModel)
```

```
##
## Call:
## lm(formula = attendance ~ population + MedHouseIncome + teamPayroll +
##     win_pct + HR + SHO, data = teams)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1350170  -371747   -37637   366870  1303918
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)     4.814e+05  1.725e+05   2.791 0.005437 **
## population      1.821e-02  5.217e-03   3.491 0.000518 ***
## MedHouseIncome -4.349e+00  1.530e+00  -2.842 0.004649 **
## teamPayroll     7.589e-03  5.793e-04  13.100  < 2e-16 ***
## win_pct         4.029e+06  3.886e+05  10.368  < 2e-16 ***
## HR             -2.493e+03  6.953e+02  -3.586 0.000364 ***
## SHO            -2.217e+04  6.690e+03  -3.315 0.000976 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 498100 on 565 degrees of freedom
## Multiple R-squared:  0.4948, Adjusted R-squared:  0.4895
## F-statistic: 92.24 on 6 and 565 DF,  p-value: < 2.2e-16
```

Every variable in this model is significant, and the model accounts for about 49% of the variation in season attendance.

Predicting Charlottes Attendance

To predict the season attendance for a potential Charlotte MLB team, we will plug in the population and median household income data, then average the other values from the dataset to see how many people an average performing Charlotte team could pull.

```r
charlotte_data <- data.frame(population = 2367000, MedHouseIncome = 84796, teamPayroll = mean(teams$tea
charlotte_data
```

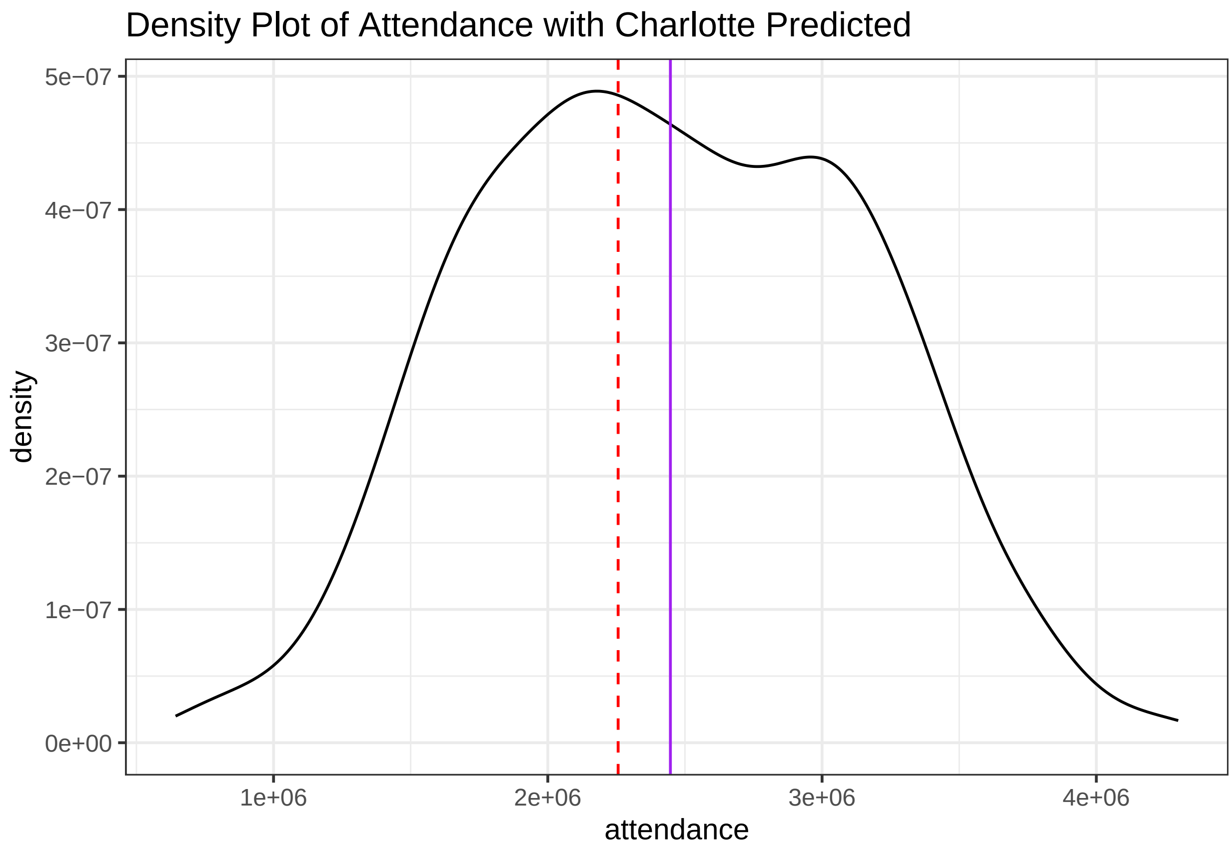
```
##   population MedHouseIncome teamPayroll   win_pct       HR      SHO
## 1    2367000          84796    95184466 0.5005305 172.0122 9.444056
```

```r
predicted_attendance <- predict(attendanceModel, newdata = charlotte_data)
predicted_attendance
```

```
##       1
## 2256422
```

The model predicts that an average performing Charlotte MLB team would pull 2,256,422 people across an entire season. This is slightly lower than the average attendance shown in the graph below with the red line indicating Charlotte's predicted attendance and the purple line indicating the league's average attendance.

```r
ggplot(teams, aes(x = attendance)) +
  geom_density() +
  geom_vline(xintercept = 2256422, color = "red", linetype = "dashed") + geom_vline(xintercept = mean(t
  labs(title = "Density Plot of Attendance with Charlotte Predicted") + theme_bw()
```



Data sources:

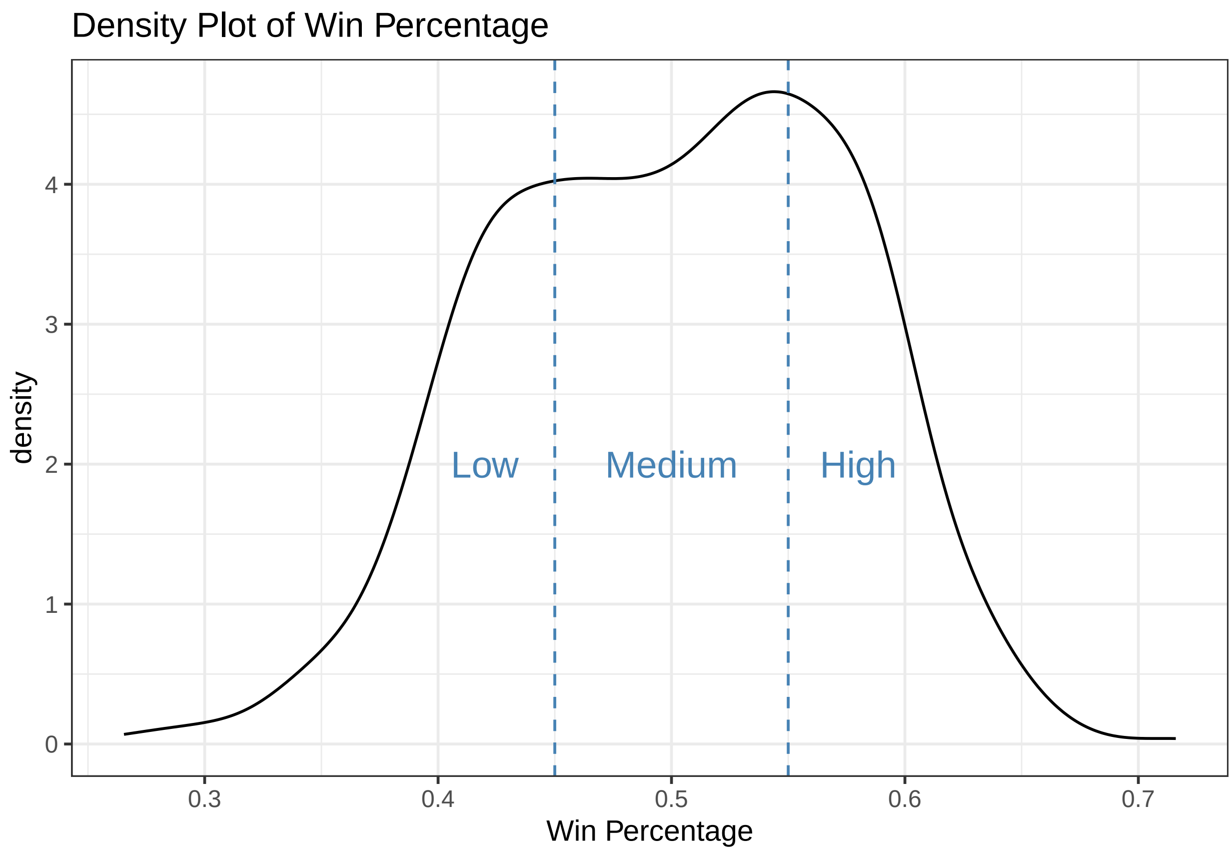
- MLB data: Lahman R package

- Population data: U.S. Census Bureau and Statistics Canada for Toronto

b)

Determining Criteria for Team Performance

```
ggplot(teams, aes(x = win_pct)) + geom_density() + theme_bw() + geom_vline(xintercept = .45, color = "s
```

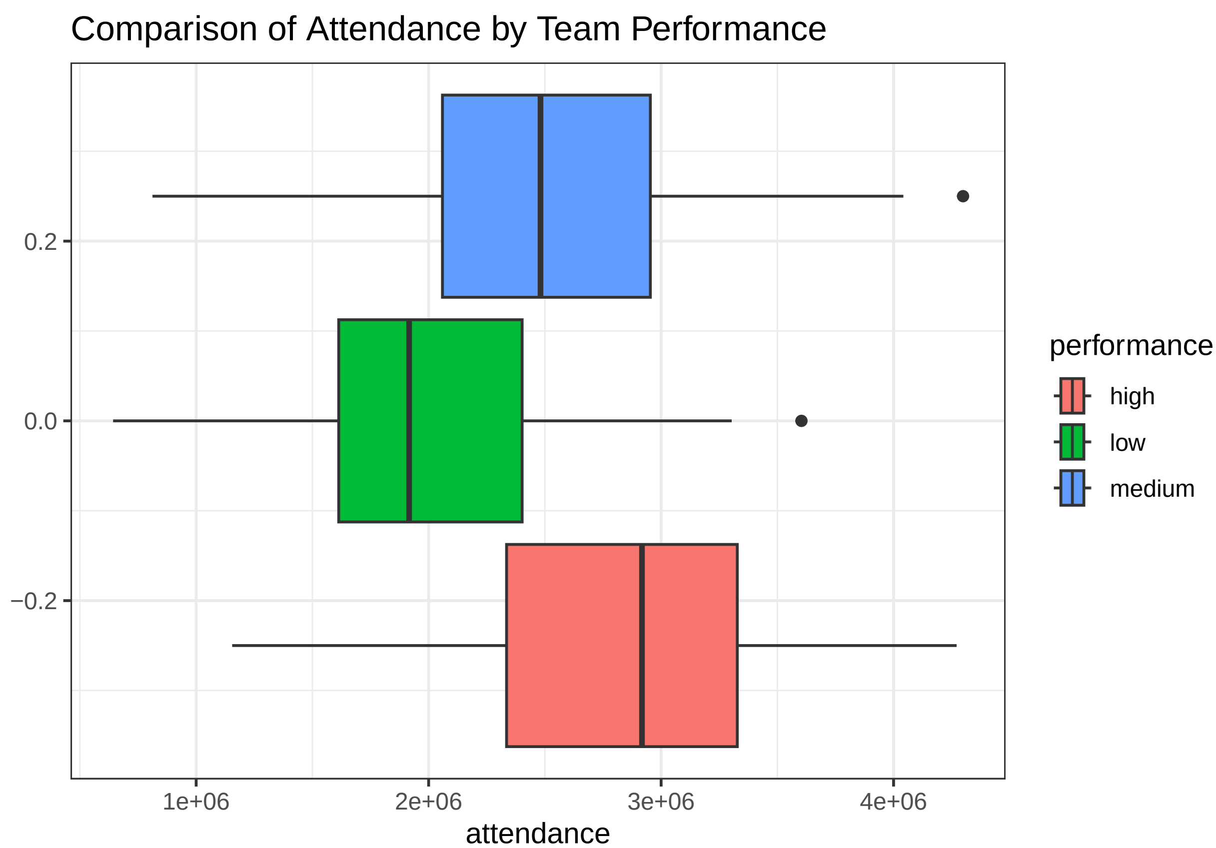


Based on the roughly normal distribution, I'll define low performance to be any win percentage less than .45, medium performance to be in between .45 and .55, and high performance to be any win percentage greater than .55.

```
teams <- teams %>%
  mutate(performance = case_when(
    win_pct <= .45 ~ "low",
```

```
    win_pct <= .55 ~ "medium",
    win_pct > .55 ~ "high"
  ))
```

```
ggplot(teams, aes(x = attendance, fill = performance)) + geom_boxplot() + theme_bw() + labs(title = "Co
```



While the graph shows overlap between the three performance categories, it seems that the median attendance values are in order of low, medium, and high attendance. We will use an ANOVA test to confirm the result.

```
anova_result <- aov(attendance ~ performance, data = teams)
summary(anova_result)
```

```
##              Df    Sum Sq   Mean Sq F value Pr(>F)
## performance   2 5.326e+13 2.663e+13   67.56 <2e-16 ***
## Residuals   569 2.243e+14 3.942e+11
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The ANOVA test has a low p-value, confirming the attendance is significantly different between levels of performance. We'll now rerun the model, replacing win percentage with performance to judge the affect of low, medium, and high performance in the context of the model.

```r
attendanceModel <- lm(attendance ~ population + MedHouseIncome + teamPayroll + performance + HR + SHO, d

summary(attendanceModel)
```

```
##
## Call:
## lm(formula = attendance ~ population + MedHouseIncome + teamPayroll +
##     performance + HR + SHO, data = teams)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1374630  -371342   -50384   408402  1401327
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)        2.527e+06  1.842e+05  13.718  < 2e-16 ***
## population         2.051e-02  5.380e-03   3.812 0.000153 ***
## MedHouseIncome    -4.676e+00  1.579e+00  -2.962 0.003188 **
## teamPayroll        7.675e-03  5.964e-04  12.869  < 2e-16 ***
## performancelow    -6.312e+05  7.541e+04  -8.371 4.54e-16 ***
## performancemedium -2.790e+05  5.718e+04  -4.879 1.39e-06 ***
## HR                -1.415e+03  6.917e+02  -2.046 0.041222 *
## SHO               -1.383e+04  6.767e+03  -2.044 0.041422 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 512800 on 564 degrees of freedom
## Multiple R-squared:  0.4656, Adjusted R-squared:  0.459
## F-statistic: 70.21 on 7 and 564 DF,  p-value: < 2.2e-16
```

Predicting Charlotte Attendenance with Varying Performance

```r
charlotte_data_low <- data.frame(population = 2367000, MedHouseIncome = 84796, teamPayroll = mean(teams$

charlotte_data_medium <- data.frame(population = 2367000, MedHouseIncome = 84796, teamPayroll = mean(tea

charlotte_data_high <- data.frame(population = 2367000, MedHouseIncome = 84796, teamPayroll = mean(team
```

```r
predicted_attendance_low <- predict(attendanceModel, newdata = charlotte_data_low)

predicted_attendance_medium <- predict(attendanceModel, newdata = charlotte_data_medium)

predicted_attendance_high <- predict(attendanceModel, newdata = charlotte_data_high)

predicted_attendance_low
```

```
##       1
## 1903882
```

```r
predicted_attendance_medium
```

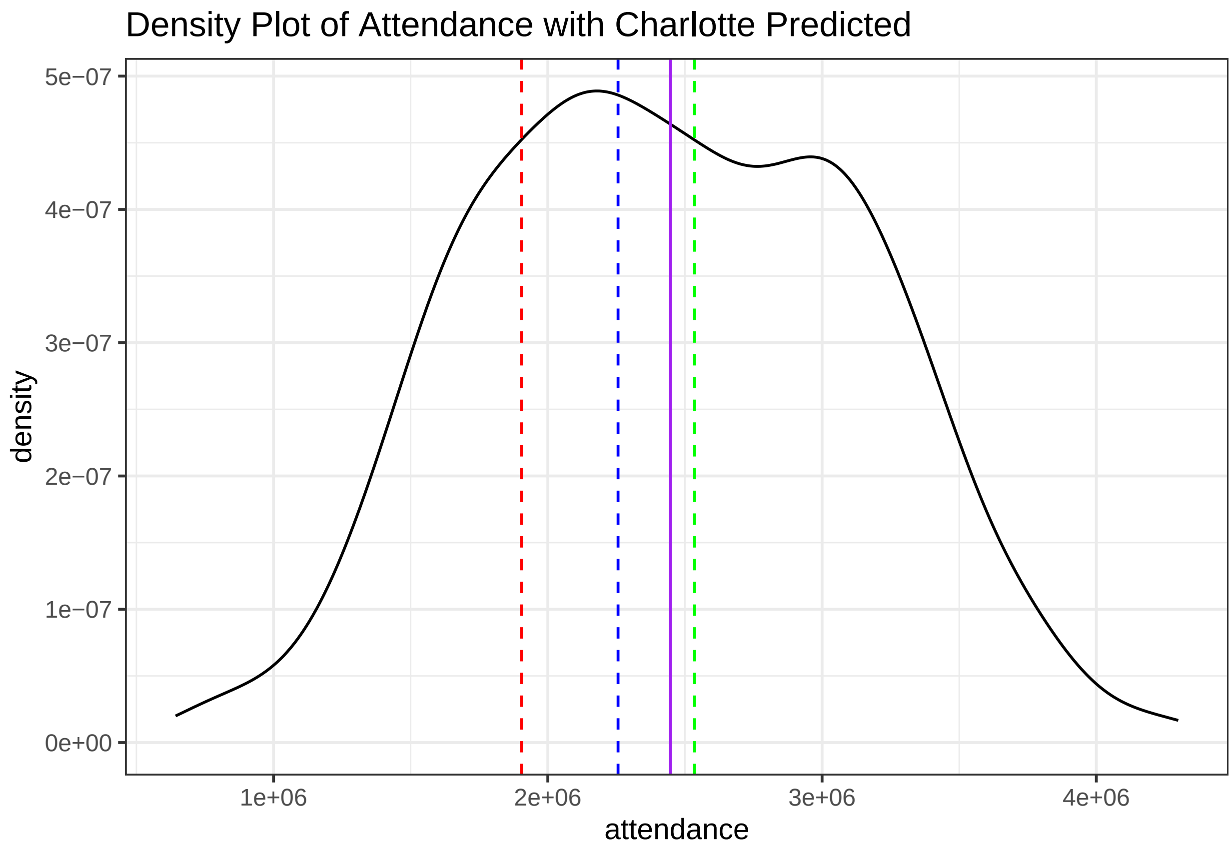
```
##       1
## 2256104
```

```r
predicted_attendance_high
```

```
##       1
## 2535080
```

Based on the model with the new performance variable, the Charlotte team with low performance is predicted to garner 1,903,882 visitors, medium performance is predicted to garner 2,256,104 visitors, and high performance is predicted to garner 2,535,080 visitors across a season. These differences are shown in the graph below with the red, blue, and green lines representing low, medium, and high performance respectively, and the purple line representing the league's average attendance.

```r
ggplot(teams, aes(x = attendance)) +
  geom_density() +
  geom_vline(xintercept = 1903882, color = "red", linetype = "dashed") + geom_vline(xintercept = 225610
  labs(title = "Density Plot of Attendance with Charlotte Predicted") + geom_vline(xintercept = mean(te
```



The graph shows that the Charlotte MLB team can only break average attendance by having high performance.

c)

c)

Create data frame with franchise ID, population, and average ticket price for available teams

```
ticket_data <- data.frame(
  franchID = c("NYA", "HOU", "CHN", "LAN", "WSN", "PIT", "CHA", "CIN", "ARI", "FLA"),
  population = c(19940274, 7796182, 9408576, 12927614, 6430489, 2429469, 9408576, 2282624, 5186958, 645
  ticket_price = c(67.75, 64.29, 55.10, 54.24, 46.02, 26.93, 25.58, 25.58, 25.15, 23.61)
)
```

Build a linear regression model for ticket price

Predictor: metropolitan population (market size influences pricing strategy)

```
model <- lm(ticket_price ~ population, data = ticket_data)
```

Summary of the model

```
summary(model)
```

```
##
## Call:
## lm(formula = ticket_price ~ population, data = ticket_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -18.7518  -7.2200  -0.7287   7.0733  23.9244
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 2.119e+01  8.061e+00   2.628   0.0303 *
## population  2.460e-06  8.388e-07   2.933   0.0189 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.18 on 8 degrees of freedom
## Multiple R-squared:  0.5181, Adjusted R-squared:  0.4578
## F-statistic:   8.6 on 1 and 8 DF,  p-value: 0.01892
```

To recommend an average ticket price for a Charlotte team

Charlotte metro population: 2,805,115 (2023 estimate from U.S. Census Bureau)

```
charlotte_data <- data.frame(population = 2805115)
```

Predict average ticket price

```
predicted_price <- predict(model, newdata = charlotte_data)
```

Output the recommendation

```
cat("Recommended average ticket price per game for Charlotte MLB team: $", round(predicted_price, 2), "
```

```
## Recommended average ticket price per game for Charlotte MLB team: $ 28.09
```

Reasoning: Ticket prices are modeled based on market size (population), as larger markets can support higher prices due to demand.

For Charlotte (mid-sized market similar to Pittsburgh or Cincinnati), the predicted price is reasonable compared to the MLB overall average of ~$38 (Statista, 2024).

Adjust based on factors like stadium design, local economy, or competitive landscape (e.g., nearby Atlanta Braves).

Data sources:

- Ticket prices: DrRoto.com for 2025 MLB average ticket prices (selected teams)

- Population data: U.S. Census Bureau and Statistics Canada for Toronto

Note: Since full list of ticket prices by team is not freely available, we use available data from top and bottom priced teams to build a model.

For a more comprehensive model, additional data could be sourced from Team Marketing Report's Fan Cost Index (paid).

d)

MLB Revenue Sources

> From Sportico Research, the MLB makes 31% of its revenue from seating sales, 26% from national revenue including shared ticket revenue, 23% from local media, 11% from team sponsorships, and 10% from concessions and parking.
>
> Source: https://www.reddit.com/r/MLS/comments/1asj5x6/sportico_revenue_breakdown_and_revenue/#lightbox

e)

Estimating Charlotte's Revenue

Assuming medium performance, Charlotte is expected to get 2,256,104 visitors throughout a season. At our estimated ticket price of 28.09, this equates to $63,373,961 in revenue from seating sales. MLB teams must pay 48% of their ticket revenue to a national fund, and receive $118,000,000 in ticket revenue shares back, as well as $91,000,000 from a national fund. This means Charlotte would pay out $30,419,501 to the national share, and receive back $209,000,000. This equates to $241,954,460 in revenue from the national fund and ticket revenue. For the average MLB team, these categories leave 43% of revenue sources unnacounted for. Dividing the revenue by .57 accounts for this, and estimates a total of $424,481,509 in total revenue for Charlotte.

From Spotrac, this revenue is in line with what other teams receive. Other MLB teams spend anywhere from 27% to 90% of their revenue on payroll. For Charlotte, that would be a range of $114,610,007 to $382,033,358 to spend on payroll while still being profitable by the standards of other MLB teams. This range puts Charlotte in the spending range of other MLB teams, meaning Charlotte could be a viable place for the next MLB team.

Sources: https://twinstrivia.com/2025/04/05/mlb-teams-revenue-versus-payroll/

https://www.blessyouboys.com/2021/12/16/22831008/mlbs-revenue-sharing-problem-and-how-to-solve-it